

Flight Deck Sentry — Tester Guide

Companion app for the DJI RC Plus · version 0.4.5 · runs in the background while you fly · alerts you to manned aircraft before they get close

What changed in 0.4.5 (the pop-ups only; when and how Sentry alerts is unchanged)

- **Every pop-up is a dark card.** "Bound aircraft lost / back", "Internet offline / back", "Sentry restarted", "Sentry sounds on", the pre-flight result, the status notification (Disarm · Open Sentry) and the update notice used Android's white notification. They now use the same dark card as the traffic banners: **amber** when something is lost, **green** when it is back, **grey-blue** for everything else. Title and one line on the pop-up; pull the shade down for the whole message.
- **The "Clear: move ..." hint holds steady.** It could flip for a second ("move NW" → "move SE" → "move NW") before the aircraft had passed. The hint now uses where your drone will be when the aircraft passes closest, and a new direction must hold for 5 s (or the aircraft must turn more than 45°) before the banner changes it.

1. What it does (and doesn't)

Sentry watches the airspace around **your drone** and **notifies you** on the controller: a sound, a vibration, and a banner over the flight app. It does not show a map and does not deconflict for you. When it fires, you look at your map or AirSense and decide. It never takes the screen while you're flying, never installs updates while armed, and only ducks the flight app's audio for the second a tone plays.

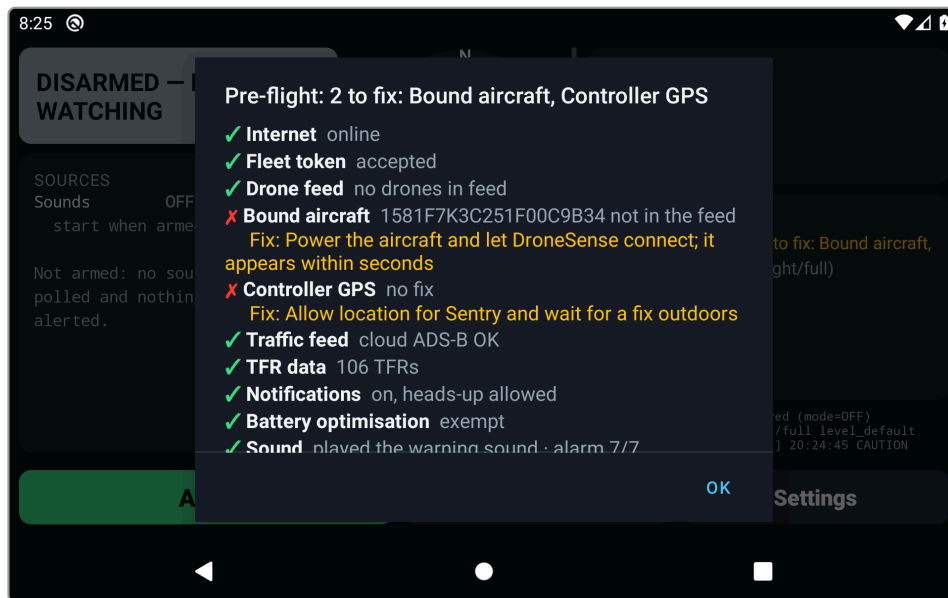
It knows where your drone is from your organization's drone feed, and where manned aircraft are from the online ADS-B feed. If your truck runs an **Overwatch ADS-B station** (in testing), Sentry also reads it directly over the truck Wi-Fi for sub-second local traffic; no other local receiver is supported. Without one, all traffic is from online. If the drone isn't in the feed yet (DroneSense closed, not signed in, or no mission selected), it protects the space around the controller instead.

Requirements. The controller must have an **internet connection** (cell or Wi-Fi) the whole flight: that is how Sentry gets your drone's position and the traffic. It only works while you are **flying in DroneSense** (open, signed in, mission selected); flying in any other app, your drone is not in the feed and Sentry protects the space around the controller instead. Android only (the RC Plus).

2. Install (5 minutes)

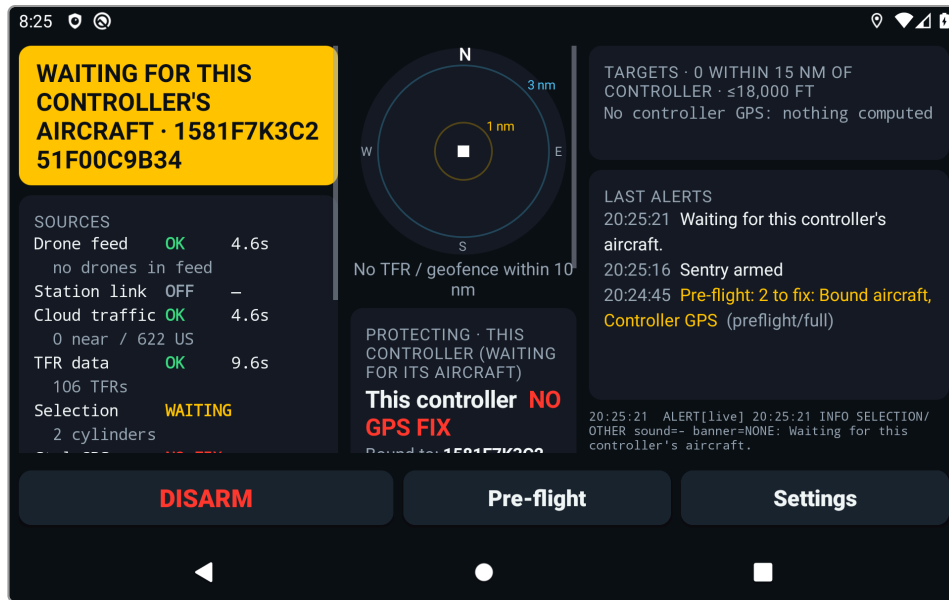
1. On the RC Plus, open the browser and go to github.com/drobinson911/flight-deck-sentry/releases/latest/download/flight-deck-sentry.apk (tap the link if you are reading this on the controller). Allow "install unknown apps" when Android asks. (If you had a version older than 0.3.0, uninstall it first.)
2. Open **Flight Deck Sentry** → **Settings**. Paste the **fleet token** (sent to you separately; never share it).

3. **This controller's aircraft: Sentry only sees your drone when DroneSense is open, you are signed in, and a mission is selected** (that is what puts it in the feed). Do that, power the drone, then tap it in the "aircraft in the feed now" list, or type its serial from the registration card. This is permanent for this controller: Sentry watches this aircraft and no other.
4. Back on the main screen tap **ARM**. Allow notifications and location, and accept the **battery-optimisation exemption** when asked. (Android's power saver freezes background apps' network and timers once the screen is off; the exemption tells Android not to do that to Sentry, so polling and alerts keep running with the screen off and DroneSense in front. Pre-flight shows red if it's missing.)
5. Tap **Pre-flight**. Fix anything with a red **X** (each line tells you how). Green all the way down = ready.
6. Switch to your flight app. Sentry keeps running; its banners appear over the flight app.



Pre-flight check: every line is a pass/fail with the fix. Here the aircraft is not powered yet and the controller has no GPS fix.

3. The main screen



Armed and waiting for the bound aircraft to appear in the feed. Left: sources and their ages. Centre: rings around the drone. Right: nearby aircraft and the last alerts. The bottom bar never moves.

- **ARM / DISARM:** Sentry only polls and alerts while armed. Once armed it **stays armed**, screen off or not, until you tap DISARM or swipe it away from the app switcher; leave it armed between hops. **Power-off:** if the controller is turned off while armed, Sentry starts itself when the controller boots (even the next day), re-arms, and posts "Sentry armed after restart"; if you had disarmed or swiped it away before power-off, it stays off. The trade-off is that an armed controller polls the feeds whenever it is on, flying or not; the cost is small (see the Resources row) but if you would rather arm by hand each day, turn off "Resume armed after power-off" in Settings. If it ever crashes, it restarts itself, re-arms, and tells you.
- **Sources:** each feed shows OK/WAIT and how old its data is. If "Drone feed" says NO TOKEN or TOKEN REJECTED, fix the token in Settings.
- **Bound to:** the serial and callsign Sentry is watching. "Waiting for this controller's aircraft" means it's not in the feed yet.

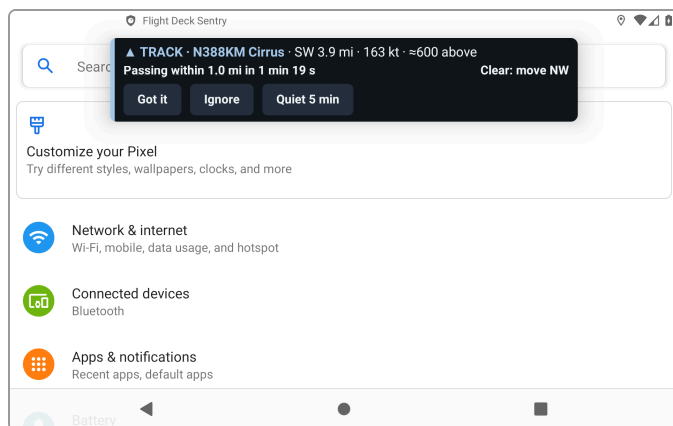
4. What the alerts mean

Alert	When	What you do
△ ADVISORY (grey-blue) / ◆ CAUTION (yellow) (ring entry)	An aircraft inside 3 nm (ADVISORY) or 1 nm (CAUTION) and inside the altitude band, not predicted to pass close. ADVISORY: one soft tone on entry, then silent banner updates. CAUTION: short sound + vibrate every 20 s while it keeps closing.	Glance at your map; keep flying.
▲ TRACK (grey-blue)	An aircraft's current track carries it through your area within 2 minutes. Popup + one sound.	Open your map, find it, start planning. Banner counts down.
⚠ WARNING (amber)	Predicted to pass within ½ mile inside 60 s, or already inside the ½-mile ring. Sound + vibrate;	Act now: move perpendicular to its track (the banner's "Clear:")

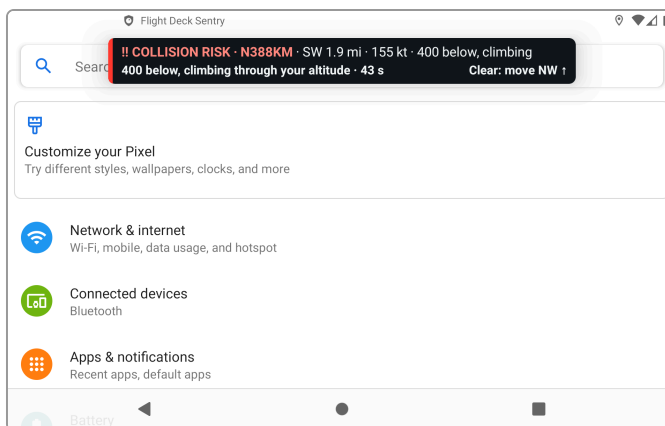
	repeats faster as it closes (every 20 → 12 → 6 s), softer on repeats.	hint), or climb/descend as advised.
!! COLLISION RISK (red)	Predicted miss under 500 ft sideways and 300 ft vertically within 60 s, or it will cross your altitude while within ½ mile. Continuous tone every 3 s. Cannot be dismissed.	Immediate evasive action.
● PASSING / ○ CLEAR (grey)	It's moving away / it's gone.	Carry on.
Zone entry	An aircraft enters the TFR you're in, or a fence you set.	Note it; the cadence above takes over.
Internet lost / regained · aircraft bound / lost	Housekeeping, one sound each.	Check the Sources panel.

Sentry alerts two ways. **Rings**: an aircraft inside a ring AND inside the altitude band (ADVISORY 3 nm, CAUTION 1 nm, WARNING ½ nm). **Prediction**: an aircraft whose speed and heading put it on course to pass close to your drone within the next 2 minutes gets a TRACK ALERT, WARNING or COLLISION RISK based on the time to that pass, **however far out it still is**; a fast aircraft 8 miles away that is 90 s from you alerts before a slow one circling at 2 miles. **Default band: surface up to 1,500 ft above your drone** (applied to where the aircraft will be at the pass, not just where it is now); adjust the ceiling in Settings to your own separation/deconfliction needs. **Every alert makes a sound** (advisory = one soft tone on entry, no repeats; warning and above repeat) so you don't have to be looking at the controller; "Quiet" style makes advisories banner-only. You never need to open Sentry while flying: stay in DroneSense and use the tools you're trained on. Sentry is a stop-gap until DroneSense warns about ADS-B traffic itself.

Which knobs to play with. The **vertical band** (ceiling above your drone) is the one meant for your ops: it shapes both the rings and the prediction, because it is applied to where the aircraft is now and to where it will be at the pass. The **rings** (3 / 1 / ½ nm) can be widened or tightened without changing the prediction: it has its own miss distances (TRACK 1 nm, WARNING ½ nm) under "Prediction" in Settings; the only crossover is that the WARNING ring is also the radius for the "crossing through my altitude" collision case, and the rings set how fast repeats come. Leave the **Prediction** section alone unless you are deliberately testing it (those are the tuned defaults below), and note anything you changed in a report.



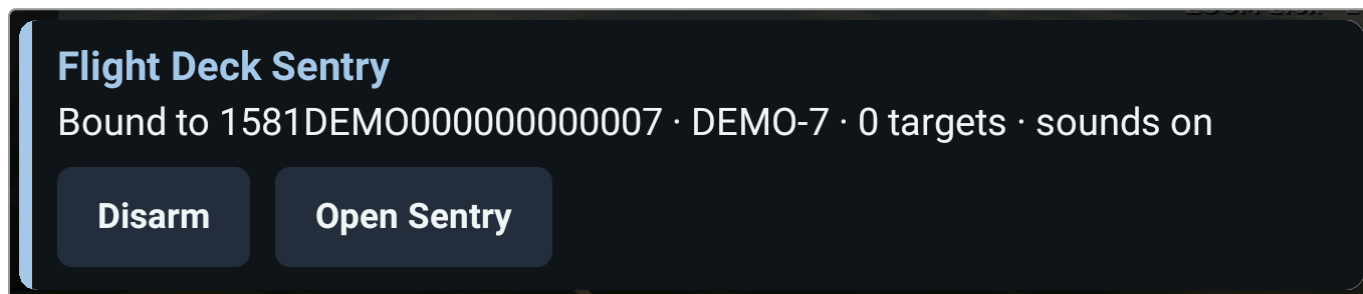
TRACK pop-up over another app: two lines and the three buttons (0.4.4).



COLLISION RISK pop-up: two lines, no dismiss buttons. Pull the shade down for all four lines.

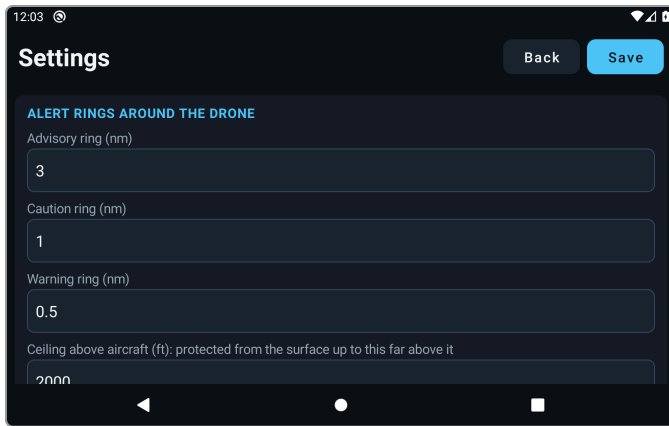
5. Banner buttons (you never need to open the app mid-flight)

- **Got it:** quiets that aircraft's repeats for 60 s. It comes straight back if the aircraft escalates, turns toward you, or becomes a collision risk.
- **Ignore:** quiets that aircraft until it clears your rings (same escape hatch). For the air attack ship you already know about.
- **Quiet 5 min:** all traffic sounds off for 5 minutes; banners keep updating silently; Sentry says "sounds on" when it ends.
- The small persistent notification has **Disarm** and **Open Sentry**; that's the only way into the app.

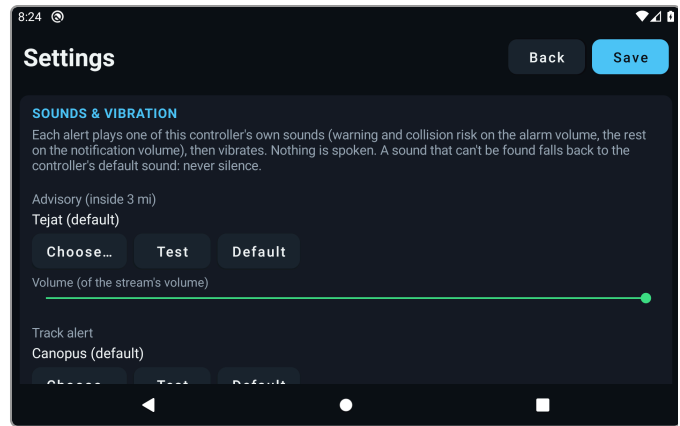


The status notification (0.4.5: its own dark card) with Disarm / Open Sentry.

6. Settings you may want to touch



Rings and the altitude band (top of Settings).



Sounds: pick any sound on the controller per alert level, Test each.

- **Rings:** 3 / 1 / 0.5 nm by default. **Ceiling above aircraft:** 1,500 ft.
- **Cylinders:** what Sentry protects around the controller when your aircraft isn't in the feed.
- **Prediction:** TRACK at 120 s, WARNING at 60 s, COLLISION RISK at 60 s to closest approach. **Why these defaults:** we replayed one full day of real drone flights (86 flights, 3,500 drone/aircraft pairs with full ADS-B tracks) through Sentry. With a 180 s track alert it produced 150 false alarms (aircraft that turned away before the predicted pass); at 120 s that fell to 32, a 79 % reduction, while warnings still came a median 69 s before closest approach and no real conflict was missed. **What adjusting does:** more seconds = more time to manoeuvre but many more false alarms (predictions 3 min out are guesses); fewer seconds = fewer false alarms but less time to react. Change them if your operation needs it, and tell us what you chose.
- **Alert style:** Standard / Quiet (Quiet doubles the intervals). **Banner duration:** 5 s.
- **Sounds & vibration:** choose per level; Test plays it. If a sound is silent, pick another.
- Every field auto-saves as you type; Enter or tapping outside closes the keyboard.

7. Try it before you fly: the demo replay

Settings → **Replay: demo encounter**. It replays a real recorded encounter (drone "DEMO-1" vs a Cirrus) through the live alert engine: you'll hear TRACK at -86 s, WARNING at -43 s, the closing repeats, PASSING and CLEAR. Tick "synthetic crossing" to hear COLLISION RISK. Nothing about the replay is live traffic.

8. Updating

Settings → **App update** shows installed vs latest and the notes; **Download and install** does the rest (one tap in Android's installer). Sentry checks once a day on its own and posts a quiet notification when there's a new one. It will never install while you're armed.

9. Test plan (you're the pros: try to break it)

Test it the way you'd test a radio you were going to bet on. For each item, note the time, what Sentry did, and what you saw on your own map. False alarms and misses are the most valuable thing you can send; "it worked" is second.

Check	How	Pass looks like
Interference with the flight app	Arm Sentry, fly a normal hop with the flight app in front. Watch video, control latency, audio.	No stutter, no dropped video, no audio loss beyond the tone's duration, no screen taken over.
Audibility and feel	Tap Test on the main screen with rotors turning and road noise. Try each level's sound in Settings.	Tone heard and vibration felt at the operating position. If not, change the sound and tell us which one worked.
Silence outside the band	Note traffic you can see on your own map that is inside 3 nm but above 1,500 ft over you (airliners, high transits).	Sentry stays silent for them. Any sound for high traffic is a bug (tell us the altitude).
Alert timing	For every TRACK/WARNING, stopwatch: banner time vs when you first saw the aircraft on your map, and how far out it was.	TRACK about 2 min out, WARNING about 60 s out, for an aircraft pointed at you. Report the numbers.
Escalation and repeats	Let a warning run without touching it.	Repeats get faster as it closes (20 → 12 → 6 s), softer than the first, stop the moment it's diverging, one PASSING, one CLEAR.
Got it / Ignore / Quiet	Use each once on a real alert.	Silence for that aircraft (or 5 min) — and it comes back on its own if the aircraft escalates or turns toward you.
Internet loss	Kill the controller's data mid-flight for 2 minutes, then restore.	One "Internet lost" alert, Sources show offline, one "regained", then normal. No flood.
Aircraft leaves the feed	Have the drone's flight app disconnect briefly, or power-cycle after landing.	"Drone position lost" → "Waiting for this controller's aircraft" → re-binds by itself when it's back.
Screen off / force-close	Turn the screen off for 10 min while armed. Separately, swipe Sentry away from Recents while armed.	Alerts still fire with the screen off. Force-closing disarms and shuts Sentry down ; it does not come back until you ARM again.
Wrong or changed aircraft	Bind the wrong serial on purpose, then fix it in Settings.	Sentry never watches another drone on its own; the fix takes

		effect live, no restart.
Pre-flight honesty	Run Pre-flight with something deliberately wrong (no token, notifications off).	The exact item shows red with a fix that works.
Battery and resources	Note controller battery at ARM and DISARM for a 30-min flight; repeat without Sentry. (The Resources row on the main screen shows CPU, memory and battery per hour.)	Report both numbers; we expect a small difference.
Two aircraft at once	When it happens, note which one Sentry talked about first.	The closer/more severe one first; the other follows with its own banner.

Send screenshots of anything confusing along with the shared log.

Bugs, false alarms and misses are the most valuable thing you can send. Settings → Diagnostics → **Share log** packages the last 24 hours of alerts, actions, feed health and resource numbers into a text file you can email from the controller.

If Sentry ever does anything dangerous, distracting, or simply not useful, say so plainly; that goes to the top of the list. Sentry is an aid, not a guarantee: it only sees aircraft that broadcast ADS-B (online, or via an Overwatch station), and it assumes they keep flying straight. See-and-avoid and your VO still come first.